

engineerHUB Presents

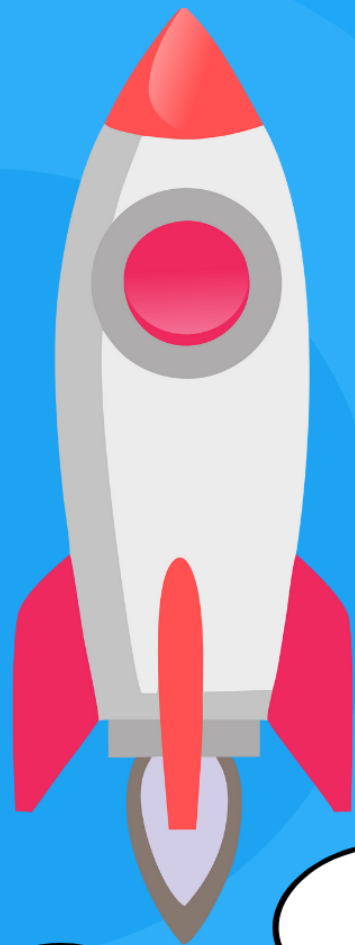
Beginners Guide to competitive programming

Competitive Programming

C#



JS





Preface

This handbook is intended to help you become a confident and successful programmer, even though you find it hard for the first time. If you follow the steps it provides, you can achieve that result. Being a successful Programmer will help you to achieve your professional dream job. It can generate personal recognition, accelerate your career, and increase your income.

People are not born with skills but the hardwork, dedication is what makes a person skillful. It is a small but very powerful handbook to give you a headstart in field of Competitive Programming.



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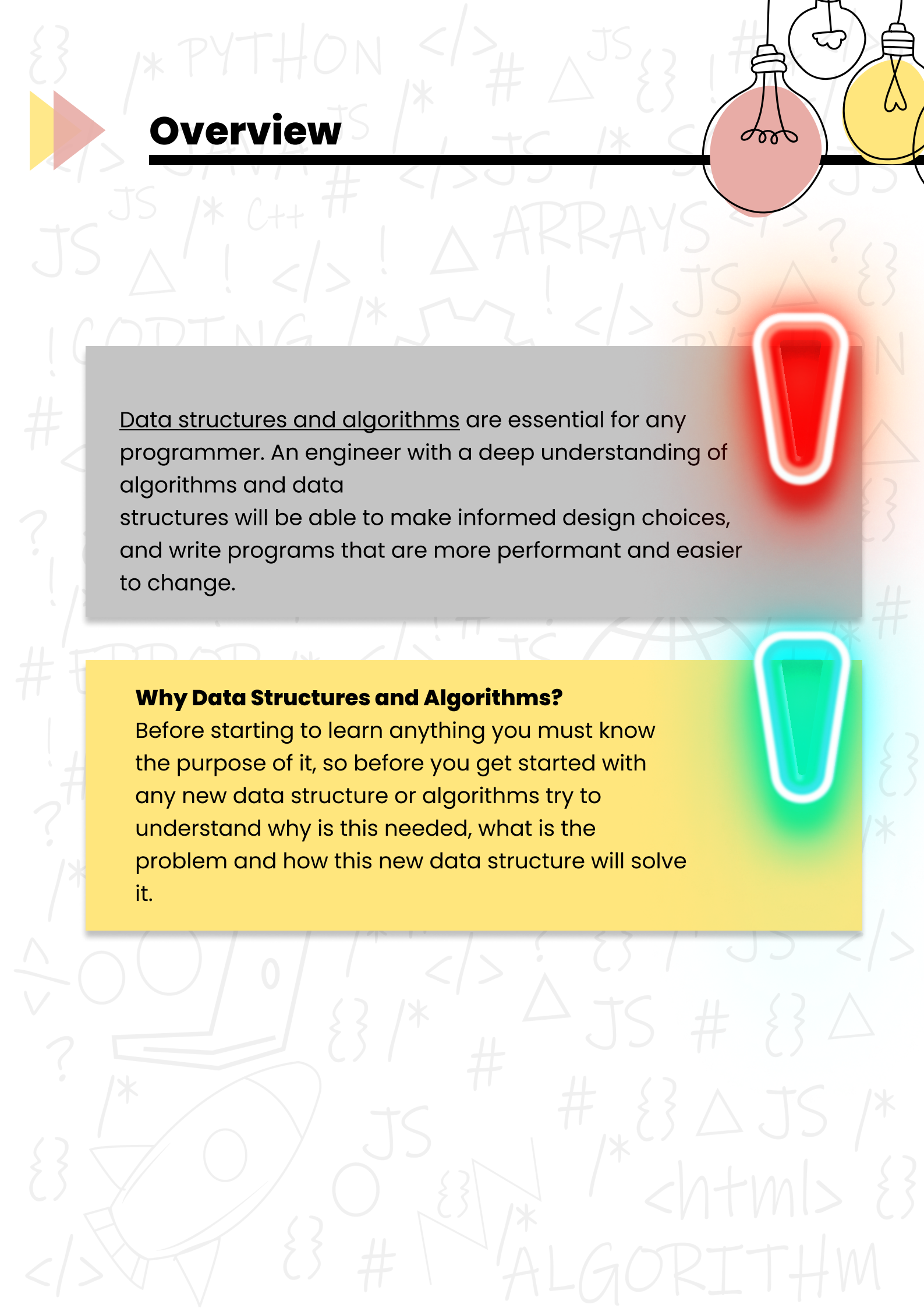
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Overview



Data structures and algorithms are essential for any programmer. An engineer with a deep understanding of algorithms and data structures will be able to make informed design choices, and write programs that are more performant and easier to change.

Why Data Structures and Algorithms?



Before starting to learn anything you must know the purpose of it, so before you get started with any new data structure or algorithms try to understand why is this needed, what is the problem and how this new data structure will solve it.

A 3D-rendered hand in a dark suit sleeve holds a glowing orange diamond-shaped sign. The sign has a bright orange border and a dark orange center. The text 'Understanding the learning Sequence' is written in white, bold, sans-serif font inside the sign. The background is solid black.

Understanding the learning Sequence



Learning Sequence

Data structures are built using a previous data structure and algorithms use the understanding of some previous algorithms as facts, so it is very important to learn things in a particular sequence. You cannot just start with trees without understanding how an array works.

1

Time and Space Complexities are Very Very Important

Whenever you are done with studying a new data structure it is very important to know the time complexities of every operation which that particular data structure performs. As every algorithm can be designed with a vast majority of data structures it is important to decide which will be the best in some particular case.

2



Learning Sequence

Competitive Programming

Competitive Programming can help you understand the use of different data structures with lots of practical problems that we commonly do not encounter even in large-sized projects. Websites like codechef, codeforces, and spoj are helping thousand of programmers to get started with understanding data structures in depth.

3

Practice, Practice and Practice

Ultimately there is no easy way to success if you want to master data structures and algorithms you need lots and lots of practice. Do not just read through internet articles explaining data structure, understanding them is very important. Code the basic structure of all the data structures. Never skip things until you understand them perfectly as they will surely haunt in the later stages of your carrier.

4

Road Map For DSA with Java



Road Map For DSA with Java

Setup

- IDE[Eclipse]
- Book: Core Java: Dr. R Nageswara Rao
- Youtube Playlist[Basic to Advance]
- Youtube Playlist[DSA and OOPS]

Platform Independency

- JVM
- Byte Code
- C++ vs Java

Learn Phase1

- Basic Syntax
- Variable
- Data Types
- Types casting

Learn Phase2

- Operators
- Conditional Statements
- Switch Statements

Advanced Java

- MultiThreading
- Exceptional Handling
- JDBC

Solving

- HackerRank

Learn Phase4

- Break in Loop
- Continue in Loop
- Taking Input using Scannner Class

Learn Phase3

- For Loop
- While Loop
- Do While Loop

Keep Practicing

Road Map For DSA with C++



Road Map For DSA with C++

Basic

- Input
- Output

Step 5

- Learning strategy of containers
- C++ (Advanced)

Syntax and terminology

- HackerRank
- HackerEarth
- W3schools
- C++

Resources

- STL Library Video Lectures
- CSES Problem Set
- HandBook
- CP Algorithms

Step 3

- Libraries

Keep Practicing

Containers

- Sequence Containers
- Associative Containers

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Resources To Learn



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Resources

Neso Academy

1) Arrays

[Play](#)

Abdul Bari Sir

2) Greedy

[Play](#)

Abdul Bari Sir

3) Hashing

[Play](#)

Aditya Verma

4) Stack

[Play](#)

Jenny's Lecture

5) Que

[Play](#)

Aditya Verma

6) Recursion

[Play](#)

Vivekananda Khayde

7) Linked list

[Play](#)

Kashishk Mehandiratta

8) Binary Trees

[Play](#)

mycode school

9) Binary Search
Tree

[Play](#)

Abdul Bari, Tech Dose

10) Strings

[Play](#)

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Resources

Abdul Bari Sir

11) BackTracking

[Play](#)

Aditya Verma

12) Binary Search

[Play](#)

Aditya Verma

13) Dynamic Programming

[Play](#)

Aditya Verma

14) Heaps

[Play](#)

Tech Dose

15) Graphs

[Play](#)

Tech Dose

16) Trees

[Play](#)

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Top 5 Online Platforms



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Platforms

01 HackerRank

HackerRank is one of the top coding practice websites out there. It is well-known for matching developers with great companies and is useful for both the employer and the employee.

Features:

- Coding practices
- interview preparations

HackerRank Login Sign Up

Matching developers with great companies.

For Companies
We are the market-leading technical interview platform to identify and hire developers wherever they are.
[Start Hiring](#)

For Developers
Join over 11 million developers, practice coding skills, prepare for interviews and get hired.
[Sign Up & Code](#)

stripe Airbnb LinkedIn Coinbase

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We use cookies to ensure you have the best browsing experience on our website. Please read our cookie policy for more information at [www.cookiepolicy.com](#)

[Start Practice](#)



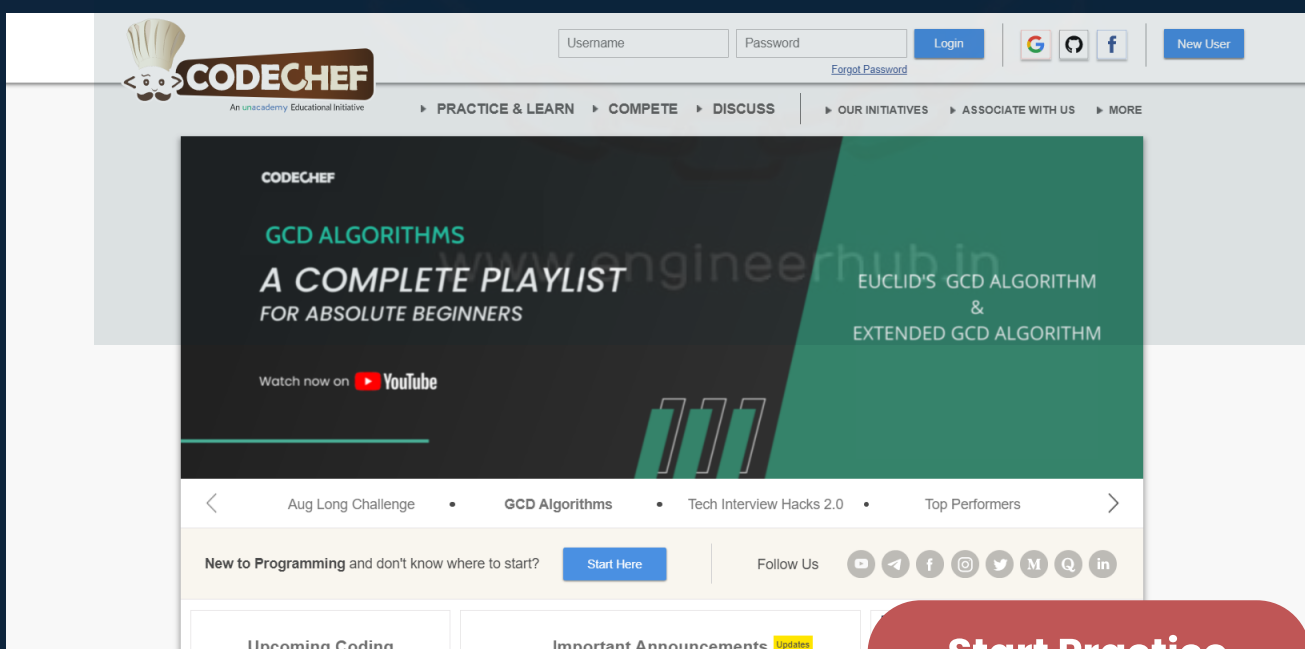
Platforms

02 Code Chef

CodeChef has is one of the biggest competitive programming platforms. They open their platform for anyone to host programming contests and a feature called campus chapter as a tool for programming clubs to utilize. They also host 3 contests each month where prizes are given.

Features :

- Practice
- Compete
- Discuss
- Host contests
- Campus chapter



[Start Practice](#)



Platforms

03 LeetCode

LeetCode has an explore tool that helps programmers progress towards the next step in their programming career. LeetCode prepares them for technical interviews, and are involved in helping companies hire top talents as well.

Features :

- Algorithms
- Database
- Shell
- Concurrency
- Contests

LeetCode Interview

Online coding interview platform for professionals

Start Interviewing

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LeetCode Interview

[Full Stack] Sam - 2 Years • Created

2020/10/10 13:23

10 Problems

Sam

Past Interviews

[Front End] Jayden - 5 Years

2020/09/28 15:20

10 Problems 1h 12m

Jayden

[Python] Cindy

2020/09/25 10:05

8 Problems

Cindy

[Start Practice](#)



Platforms

04 CodeForces

Codeforces:

CodeForces is great for programmers who seek for fun challenges to build up their skills and join groups that suit their needs. The Gym feature is also very unique where coders can train their skills in specific challenges.

Features :

- Contests
- Problem sets
- Groups

The screenshot shows the Codeforces website interface for a contest. The header includes the Codeforces logo and navigation links. The main content area displays a table of problems for the current contest.

#	Name	standard input/output	1 s, 256 MB		
A	Cubes Sorting	standard input/output	1 s, 256 MB		x5668
B	Rock and Lever	standard input/output	1 s, 256 MB		x2524
C1	Pokémon Army (easy version)	standard input/output	2 s, 256 MB		x586
C2	Pokémon Army (hard version)	standard input/output	2 s, 256 MB		x39
D	Rescue Nibel!	standard input/output	2 s, 256 MB		

Codeforces Round #672

System testing

35%

→ Contest materials

Announcement

[Start Practice](#)



Platforms

05 HackerEarth

HackerEarth is a popular coding platform that features over 8000 questions, 2000 challenges, and 1000 hackathons. It is also trusted by hiring managers across 1000+ companies. This platform has a thriving community of 3 million developers.

Features :

- Practice (basic programming questions)
- Contests (from beginners to week-long marathons)
- Hackathons (solving real-world problems and business challenges)
- Data Science competitions (testing your data science skills)

The screenshot shows the HackerEarth homepage. At the top left is the 'hackerearth' logo, and at the top right are 'Login' and 'Signup' buttons. The main heading reads 'Matching great developers with world-class companies'. Below this, there are two columns. The left column is for 'Businesses' and features 'HackerEarth for Enterprises', describing how the platform helps engage and source top developers. It includes a 'Hire now' button. The right column is for 'Developers' and features 'HackerEarth Community', encouraging users to sharpen their skills and compete in challenges. It includes a 'Join now' button. At the bottom, there are logos for 'PayPal', 'amazon', 'Walmart Labs', and 'dunnhumby'. A large red button in the bottom right corner says 'Start Practice'.

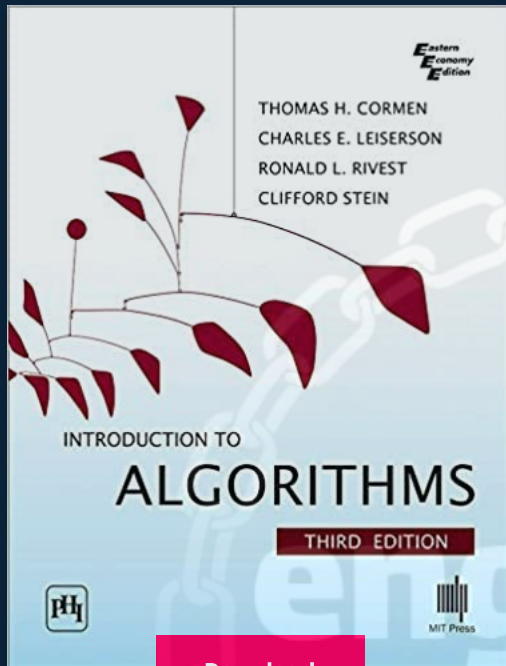
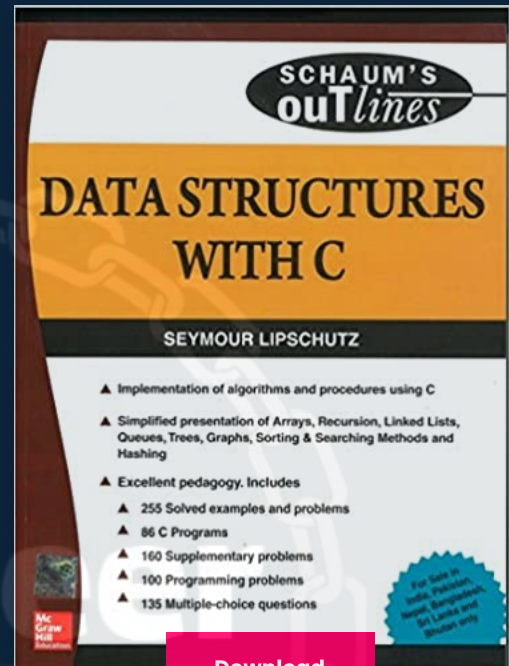
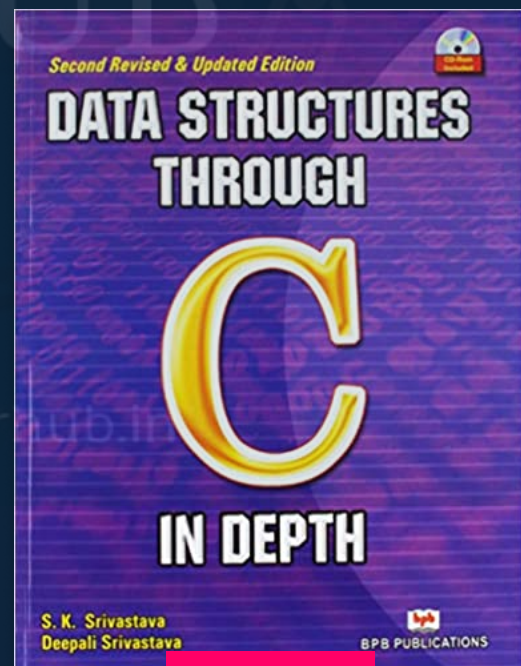
Top Books



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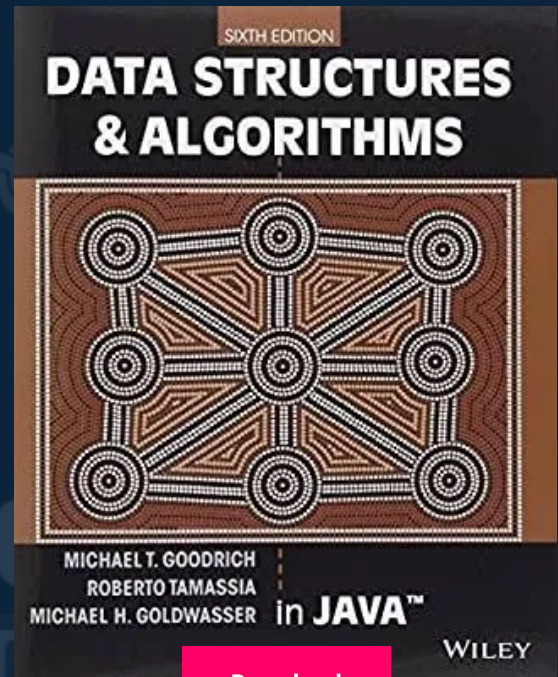
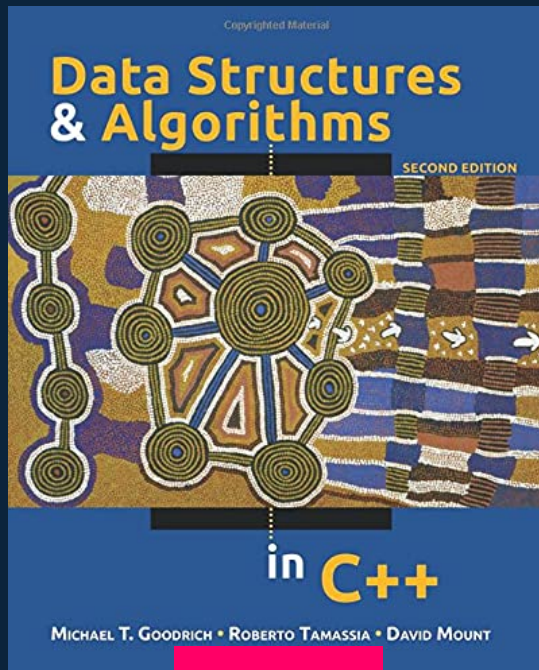


Books

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Books





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